



BAHIR DAR UNIVERSITY

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1. INTRODUCTION (BACKGROUND AND MOTIVATION)

Microsoft released Windows 11 in October 2021 as the most recent primary version of the Windows operating system because the company wanted to provide a modern and safe computing experience. The visual layout of Windows 11 is improved by a centered Start menu and Snap Layouts which are advanced window management tools because these features increase the simplicity of operation for individuals. Many people believe that Windows 11 was constructed to satisfy the requirements of modern technology while supporting new hardware and faster performance. Stronger safety is emphasized by Windows 11 through technical requirements like TPM 2.0 and Secure Boot because Windows 11 seeks to protect the information of the user. Since Windows 11 connects effectively with cloud services and working position applications, Windows 11 represents a significant step in the evolution of desktop operating systems.

BACKGROUND

Microsoft distributed Windows 10 as the primary software for a lengthy duration because the company chose to continue without alteration of this version instead of introducing a successor. Many researchers claim that the emergence of sophisticated computer equipment and growing digital safety risks compelled the corporation to create a more robust operating system. Windows 10 was maintained for many years by Microsoft while the world changed. Because technology changes quickly, Windows 10 was eventually viewed as insufficient for modern requirements. Individuals who utilize computers expressed a strong desire for a visual layout that is attractive and helpful when multiple tasks are performed simultaneously. Experts claim that the need for better cooperation through digital tools influenced the development of new software.

Since many technical advancements appeared during the recent period, Windows 11 was constructed by developers as a completely fresh start. Many experts believe that the creation of Windows 11 was necessary to match the progress of modern machinery. This Windows 11 provides a visual display that is centered in the middle of the screen so that the act of moving through the system becomes simple for every individual. While Windows 11 serves the needs of private individuals, Windows 11 also provides tools for large business organizations. Because the software is aligned with contemporary electronic components, Windows 11 is utilized effectively with new technology. It has been observed that the centered Start Menu provides a better experience for those who work on computers.

MOTIVATION

Microsoft built Windows 11 because the using computers experience needs to become superior.

- Snap Layouts and Virtual Desktops assist individuals when those individuals organize their visual working surface area. Increased work efficiency is achieved when Windows 11 manages the visual working surface area.
- Windows 11 functions with high speed while it manages information storage space well.
- Because Windows 11 saves energy, energy storage duration becomes long so that individuals finish their tasks.
- Superior functioning and working well with new hardware happen when Windows 11 is used.
- Windows 11 keeps individuals safe because complex digital attacks are stopped by the system.
- Digital safety is improved by the hardware-level protection inside Windows 11.

Due to the fact that the modern way of working is changing, Windows 11 handles these new situations.

2. OBJECTIVES

Window 11 operating system exists because a contemporary, fast, and safe digital space is needed by individuals today.

- **Improve User Experience:** Microsoft's Windows 11 shows a tidy and beautiful look. Specific parts like the new, pinned-center Start Menu and the changed taskbar were created again because ease of use helps individuals. It has been observed that the interactive parts are now very simple.
- **Enhance Productivity:** Better speed in how individuals utilize special functions is being targeted. Functions such as Snap Layouts and virtual desktops exist so that individuals can do many tasks at one time. While the window management improvements keep software in order, the working position of the user becomes easier
- **Strengthen Security:** High safety is a focus for Windows 11 because hardware-based protections like TPM (Trusted Platform Module) and secure boot are included. It is stated by many people that unauthorized entry is prevented by the system and protection is built directly into the computer parts.
- **Optimize Performance:** it focused on performance and efficiency. it facilitates quicker boot times, better management of memory, and greater power efficiency of hardware capabilities.
- **Support Modern Hardware and Applications:** Windows 11 is an operating system that works best with the latest hardware like new processors touchscreen augmentative computers and high-performance machines. It is also compatible with the most recent versions of software as well as cloud-based solutions.

3. REQUIREMENTS

To install Windows 11, a computer must meet the following minimum specifications:

➤ Hardware Requirements

Component	Minimum Requirement
Processor (CPU)	1 GHz or faster with at least 2 cores on a compatible 64-bit processor
RAM (Memory)	4 GB or more
Storage	64 GB or larger
Graphics Card	DirectX 12 or later with WDDM 2.0 driver
Display	9-inch or larger with HD resolution (720p)

➤ Software Requirements

People usually perform the setup of Windows 11 by changing their Windows 10 system or by putting the software on a matching electronic tool for the first time.

- UEFI firmware with Secure Boot must be activated by the user so that Windows 11 functions on the physical machine.
- Many experts believe that modern hardware controllers are necessary so that every physical part of the machine works with good speed.
- Windows 11 forces the person to use a Microsoft account during the configuration of the home edition because Windows 11 requires this connection to provide digital file storage and program downloads.
- A strong web link is needed while the software is being placed because Windows 11 must receive safety fixes and new options.
- Microsoft Edge is included in the software when the computer starts which allows the person to reach the internet for viewing websites and receiving help from the company.

These specific conditions allow Windows 11 to work with high performance, remains secure and provide users with access to modern features.

4. INSTALLATION STEPS

In order to install the window 11 OS, we have to use different kinds of virtualization tools and one of the most common tools for this purpose is VMware workstation pro, which allows users to create and run virtual machines on a physical computer.

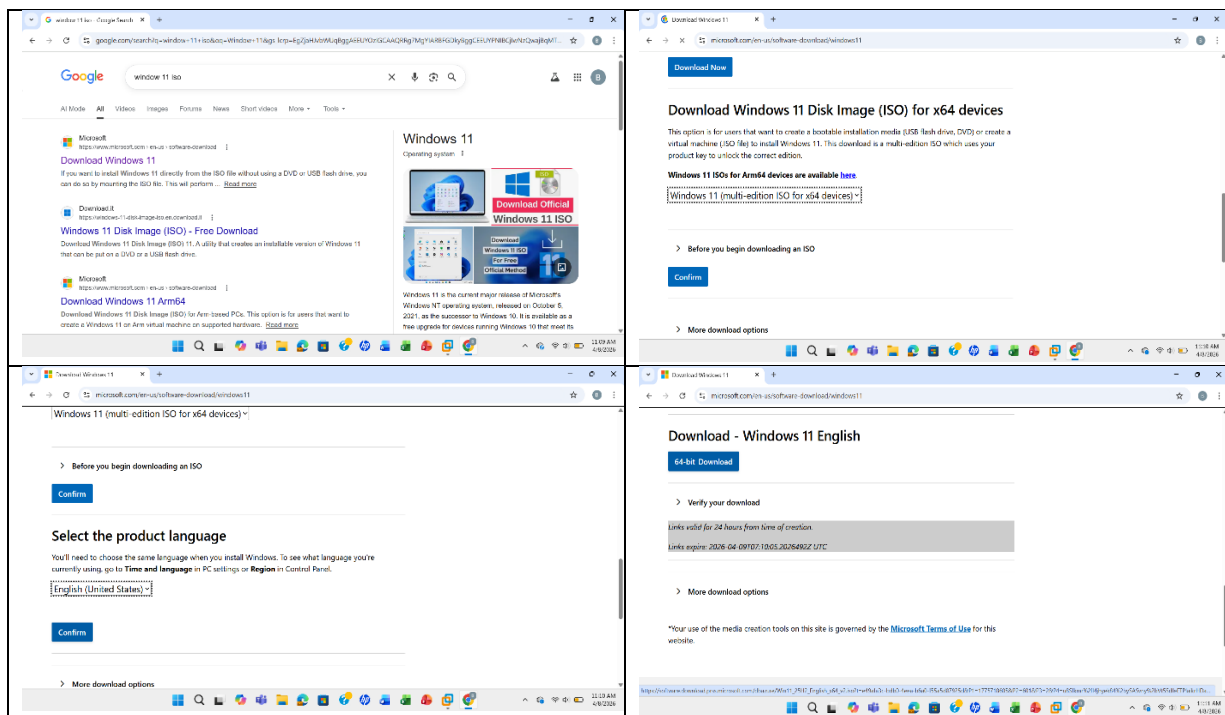
I. Installing VMware Workstation Pro

- Open the web browser and search for “**VMware Workstation download**”.
- Go to the official VMware website (from VMware)
- Download the Workstation Player installer suitable for our operating system (VMware Workstation Pro 25H2 for Windows)
- Once downloaded, execute the setup file

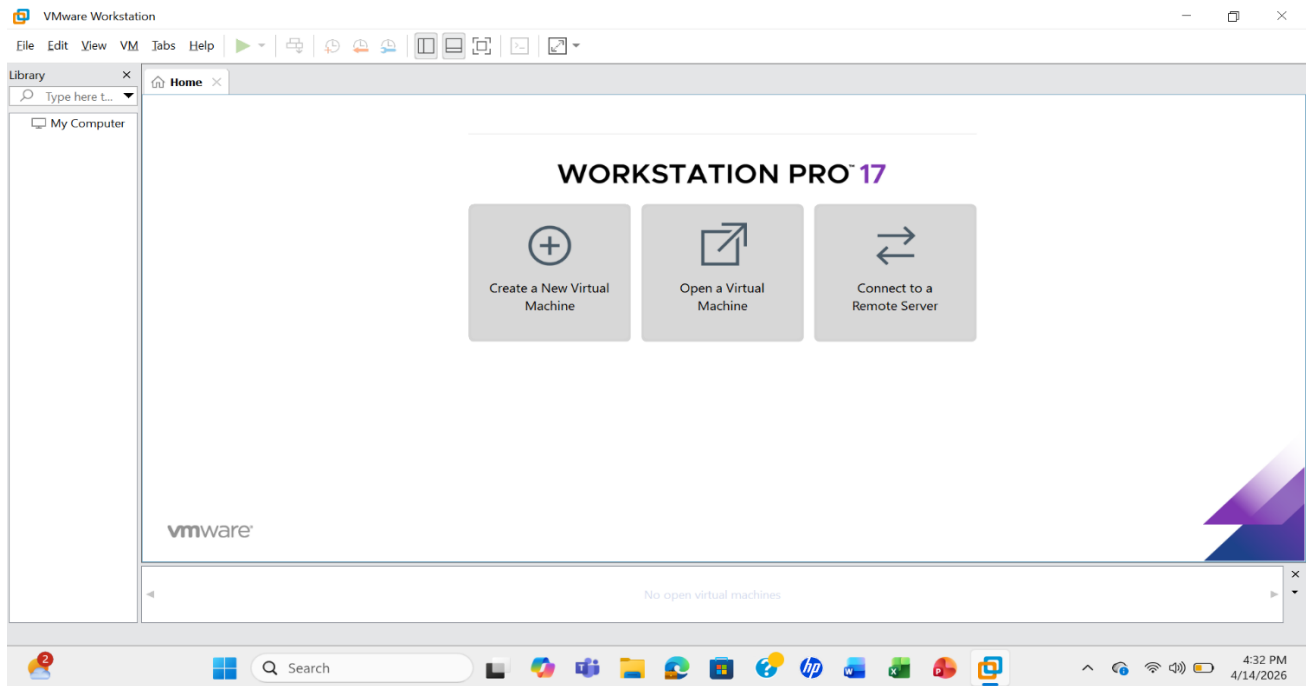
II. Installing Windows 11 on VMware Workstation Player

1) Download the Window 11 disk image file (ISO)

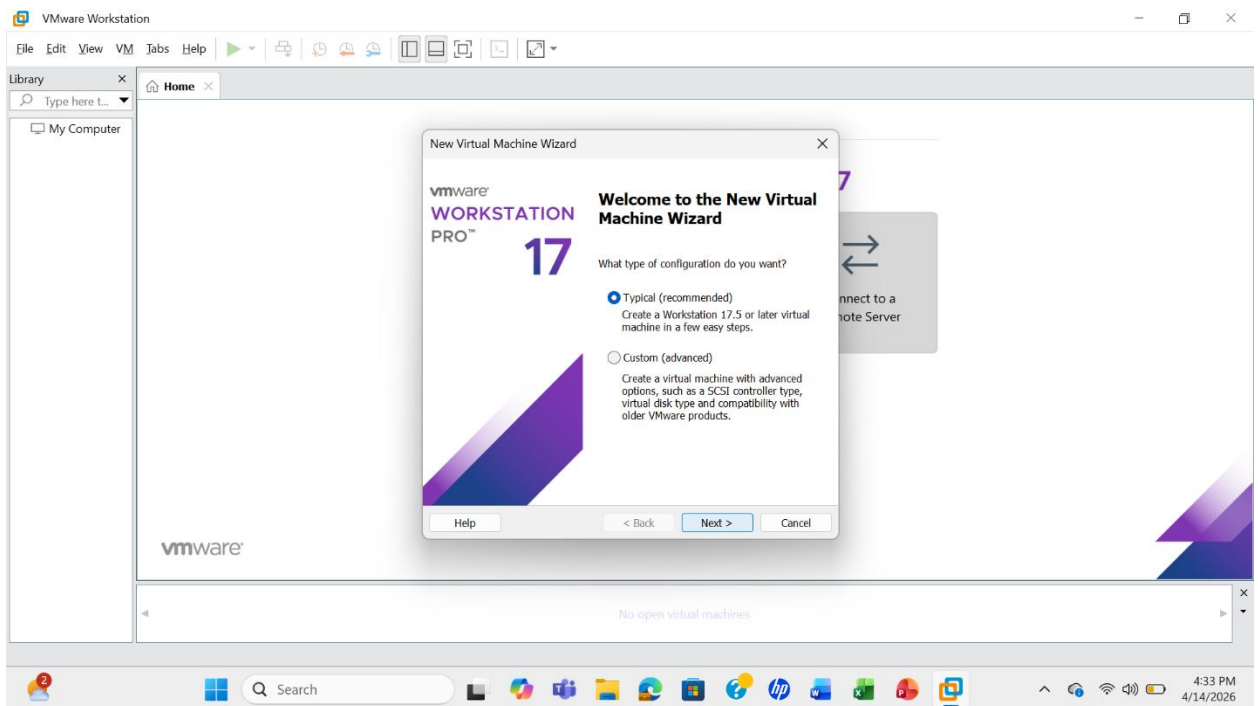
- Select the Window 11 multi-edition
- Choose the product language (English US)
- Select the 64-bit download



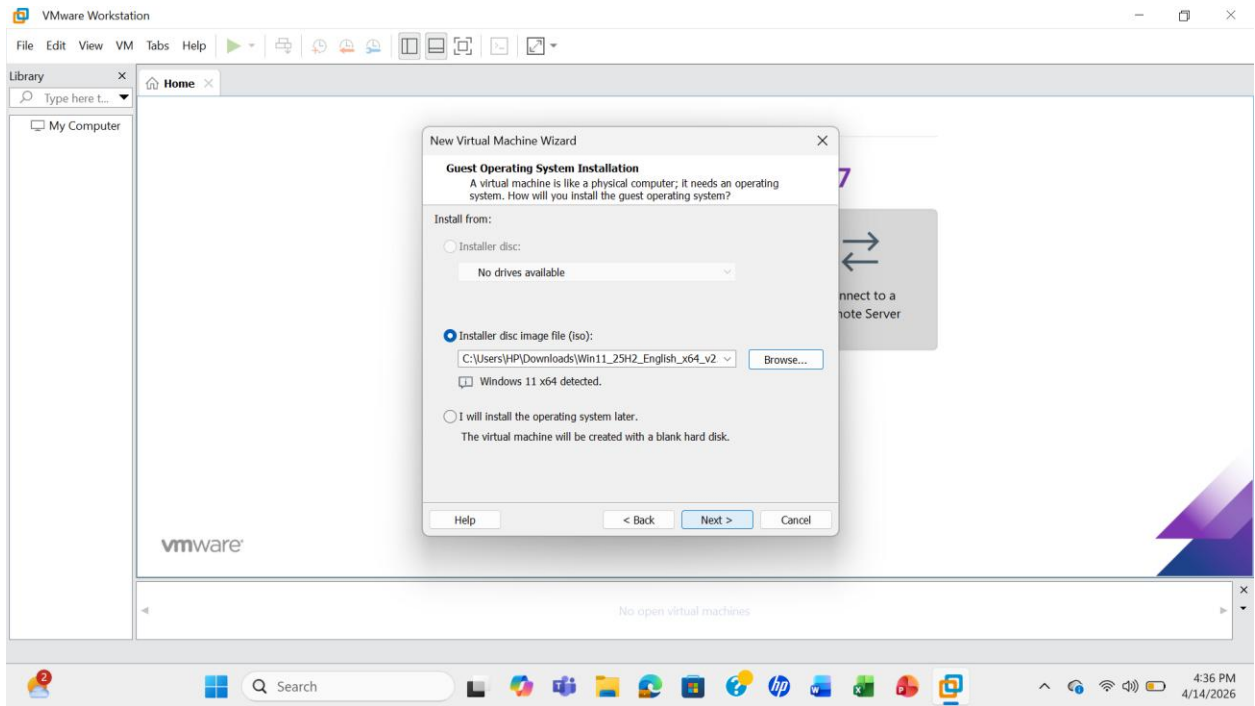
2) Open VMware Workstation Player



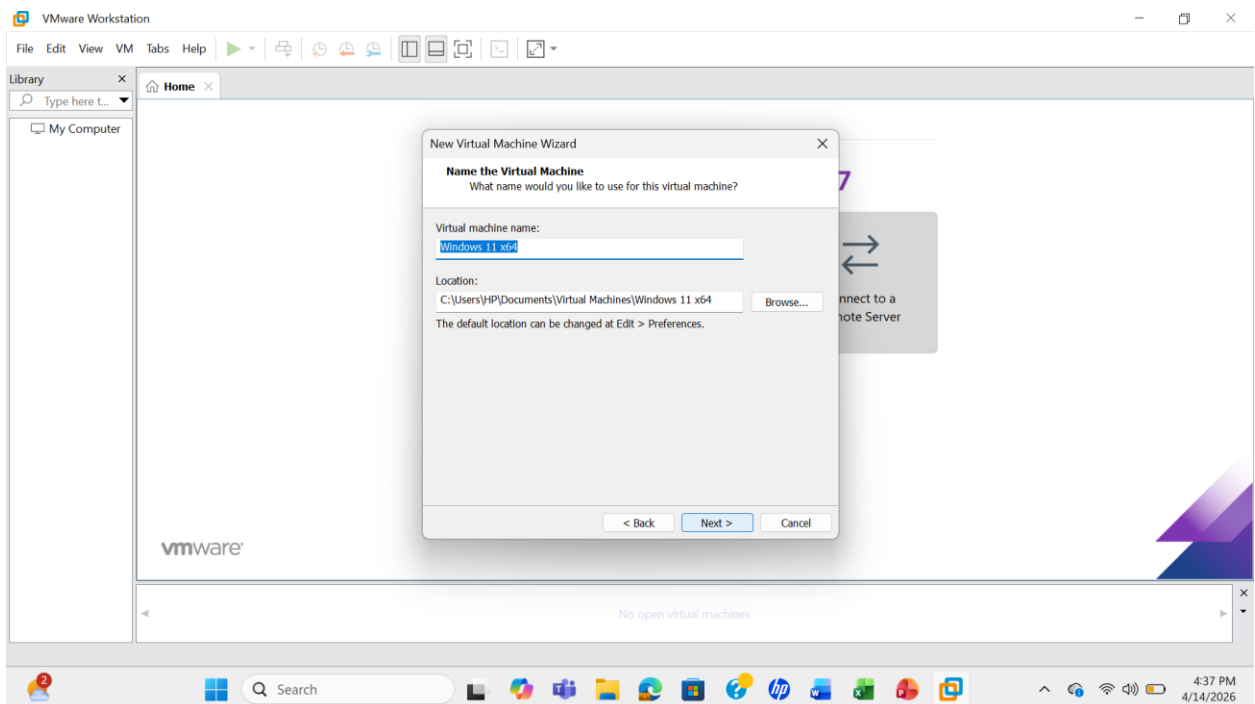
3) Select Create a New Virtual Machine and the system will open the next window. We select the typical option because provides a simple and fast installation process without requiring advanced configuration and click next.



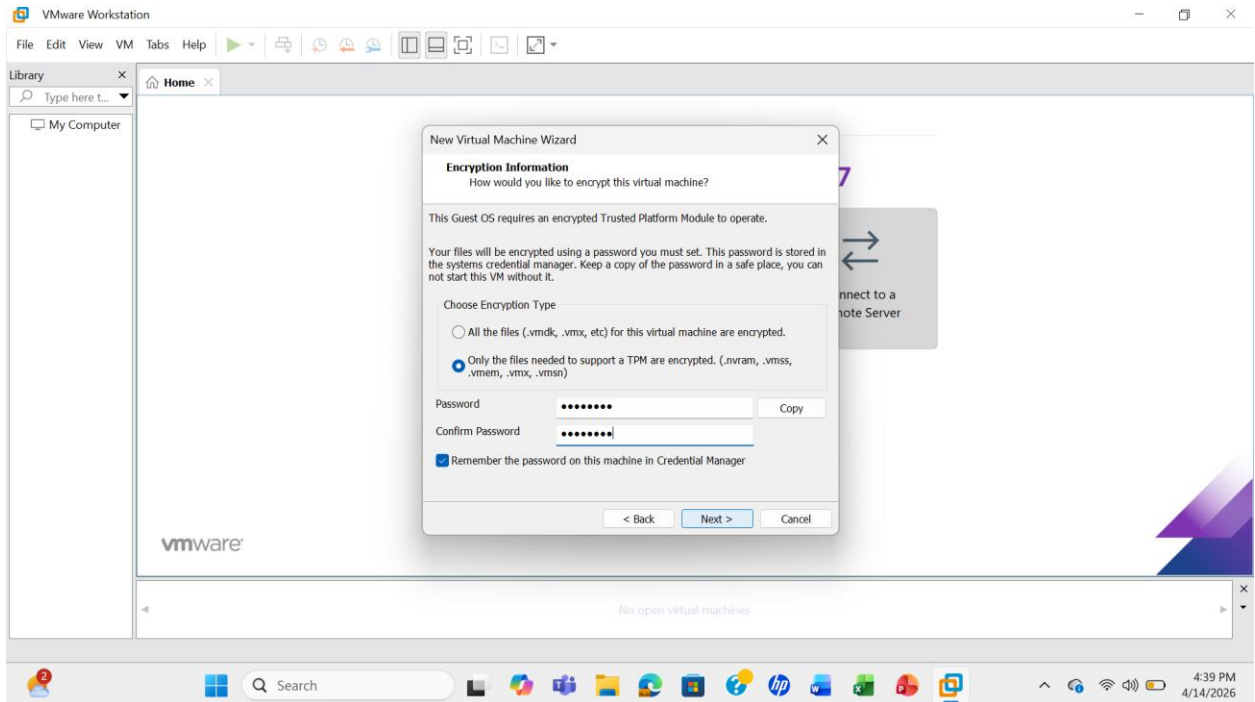
- 4) Browse and select the Windows 11 disk image file (ISO) that is already installed in our previous step and click next.



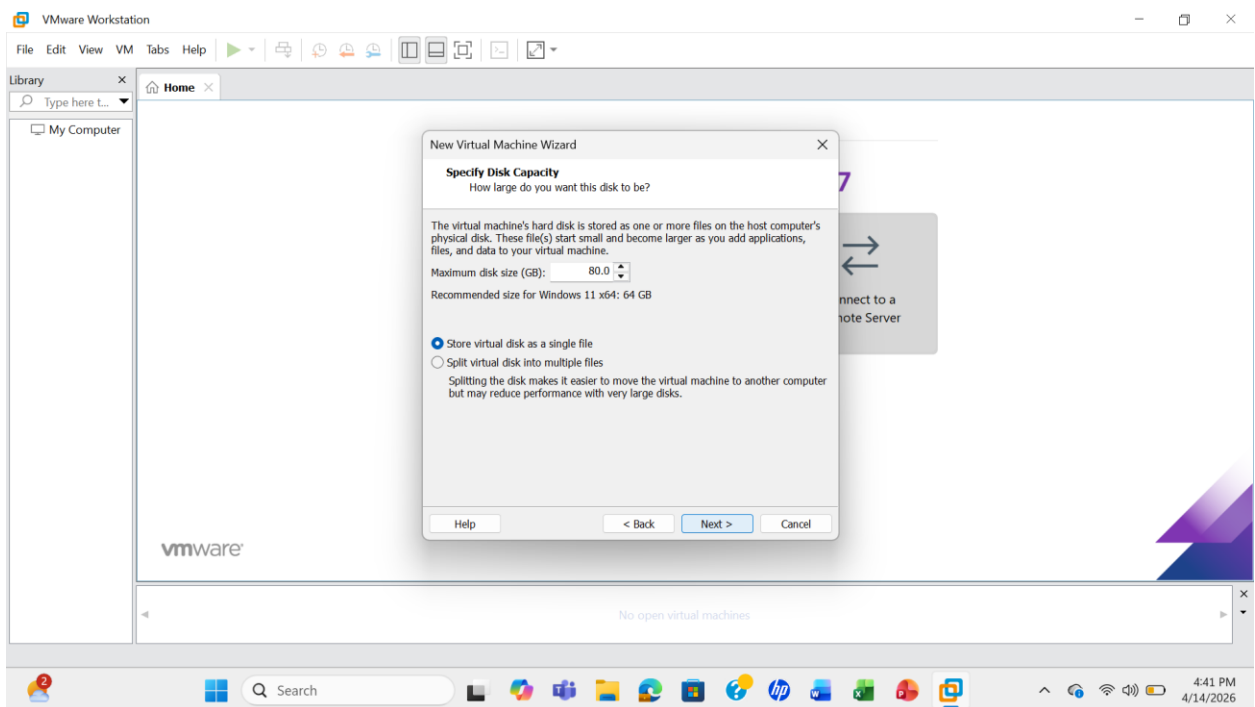
- 5) Name our virtual machine



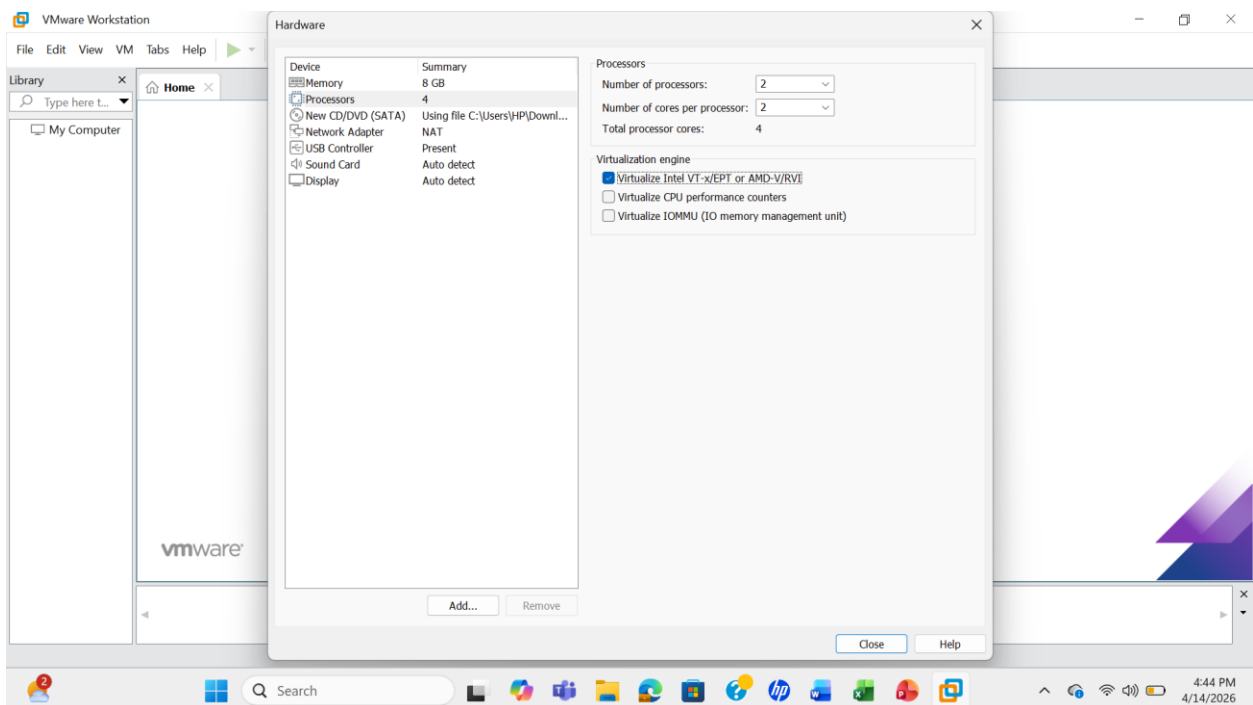
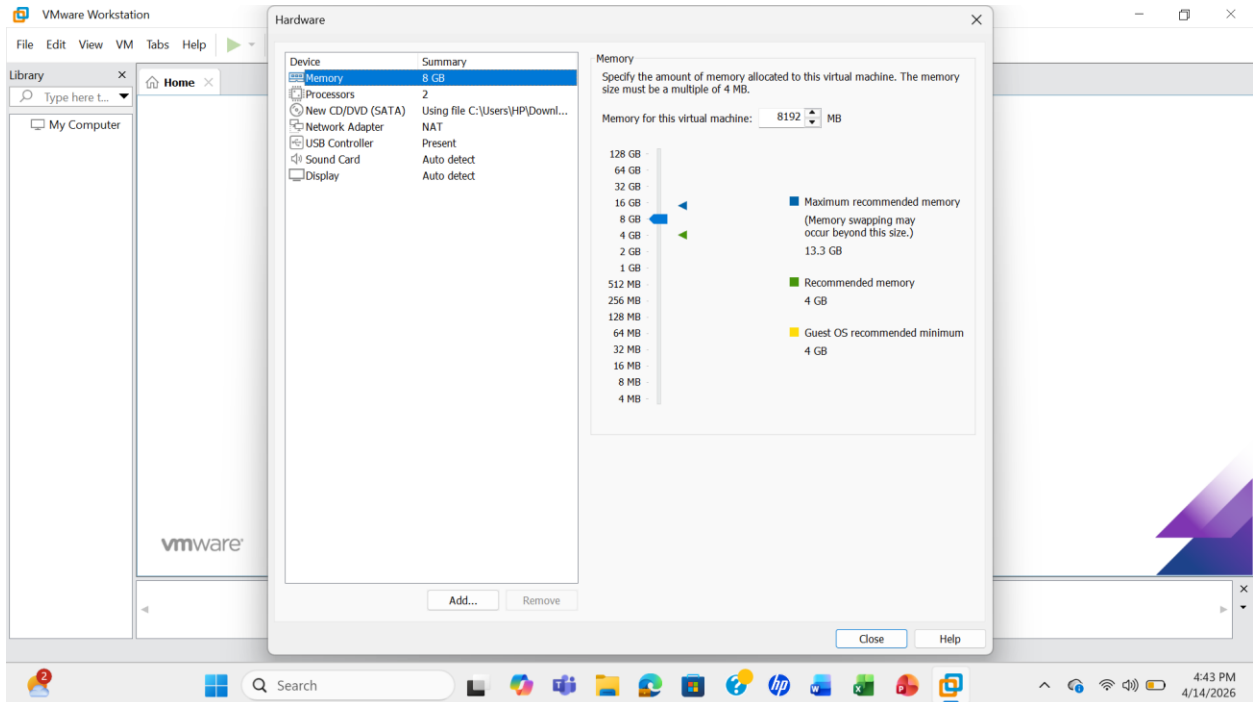
6) Choose the encryption type and set the password

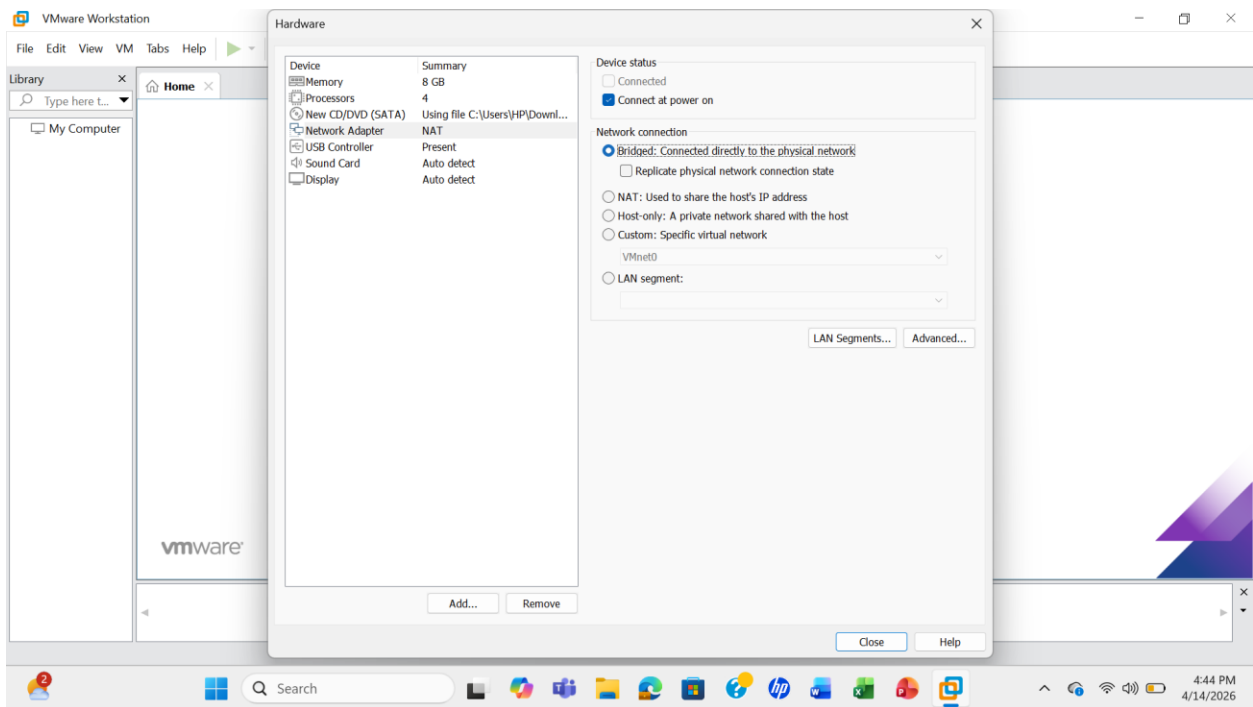


7) In the next step, we allocate system resources for the guest OS. The recommended disk size for Window 11 is 64GB but this value is not fixed and can be changed depending on the user's available storage and specific requirements. And we select store virtual disk as a single file.

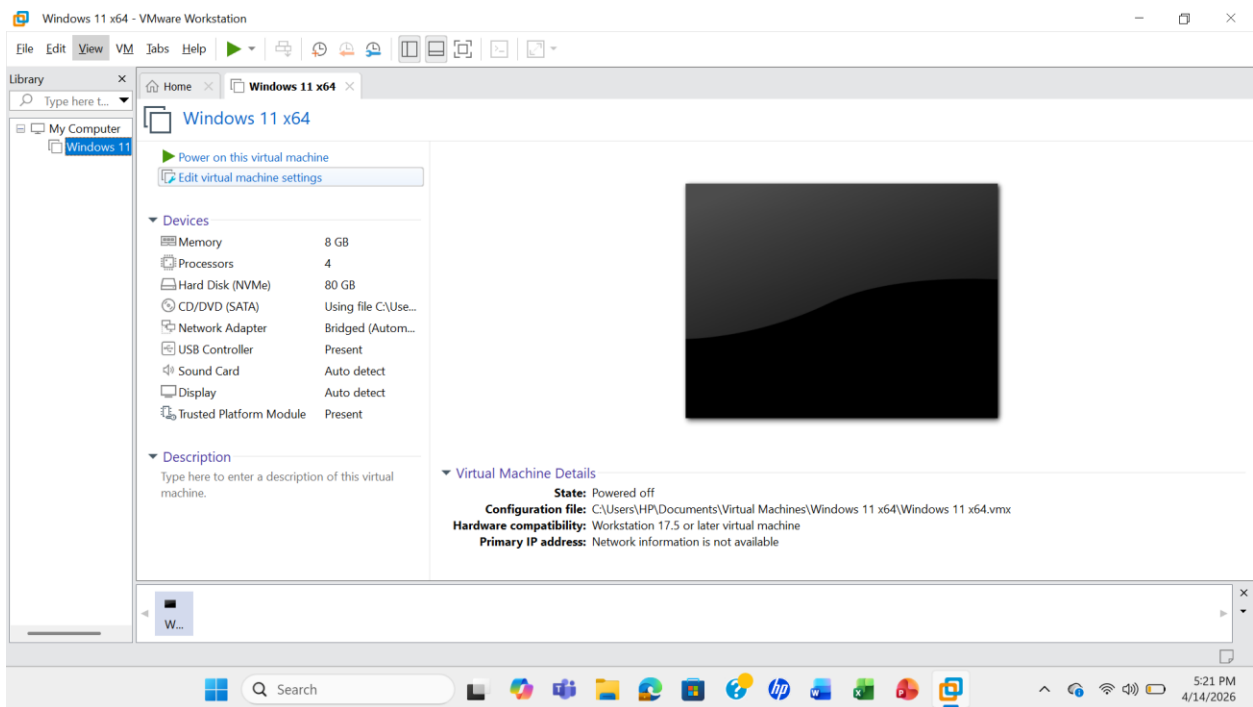


- 8) Specify the amount of memory allocated (recommended- 8GB) and the number of processors (at least 2 cores) and enable Intel Virtualization Technology (Intel VT-x) to allow the Window 11 run smoothly. For the network adapter, we chose the bridged networking because it allows the virtual machine to connect directly to the same network as the host computer.

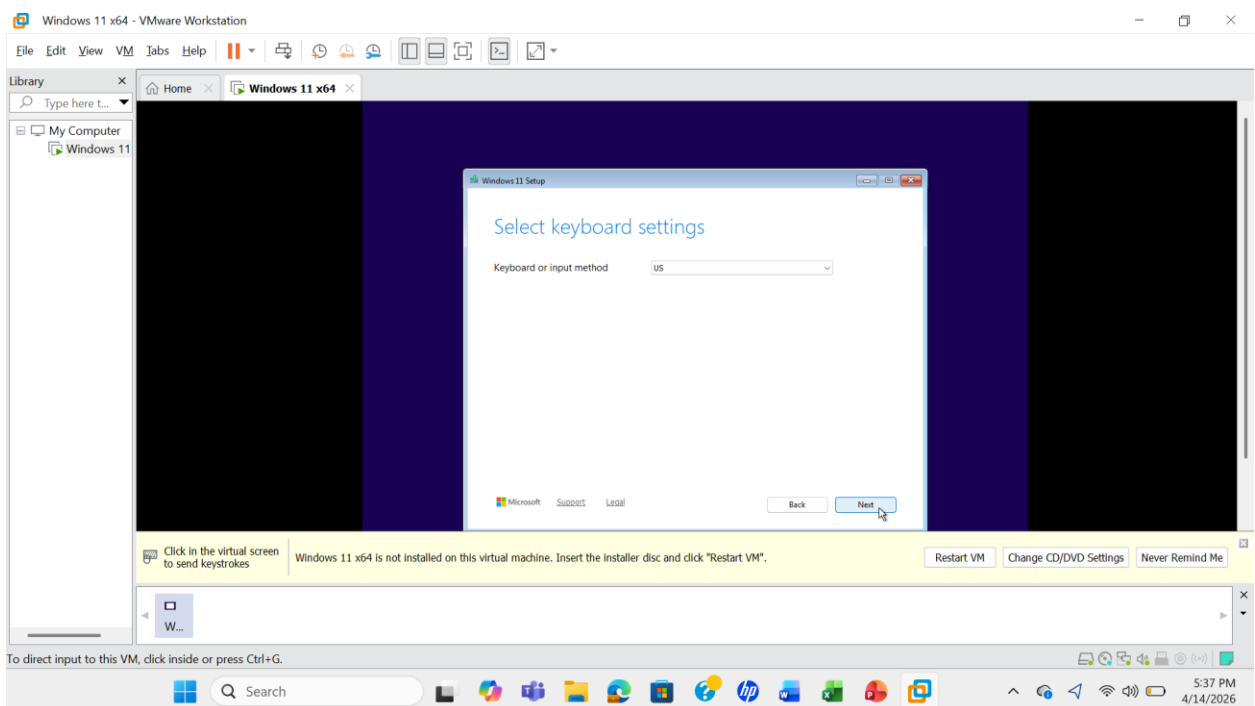
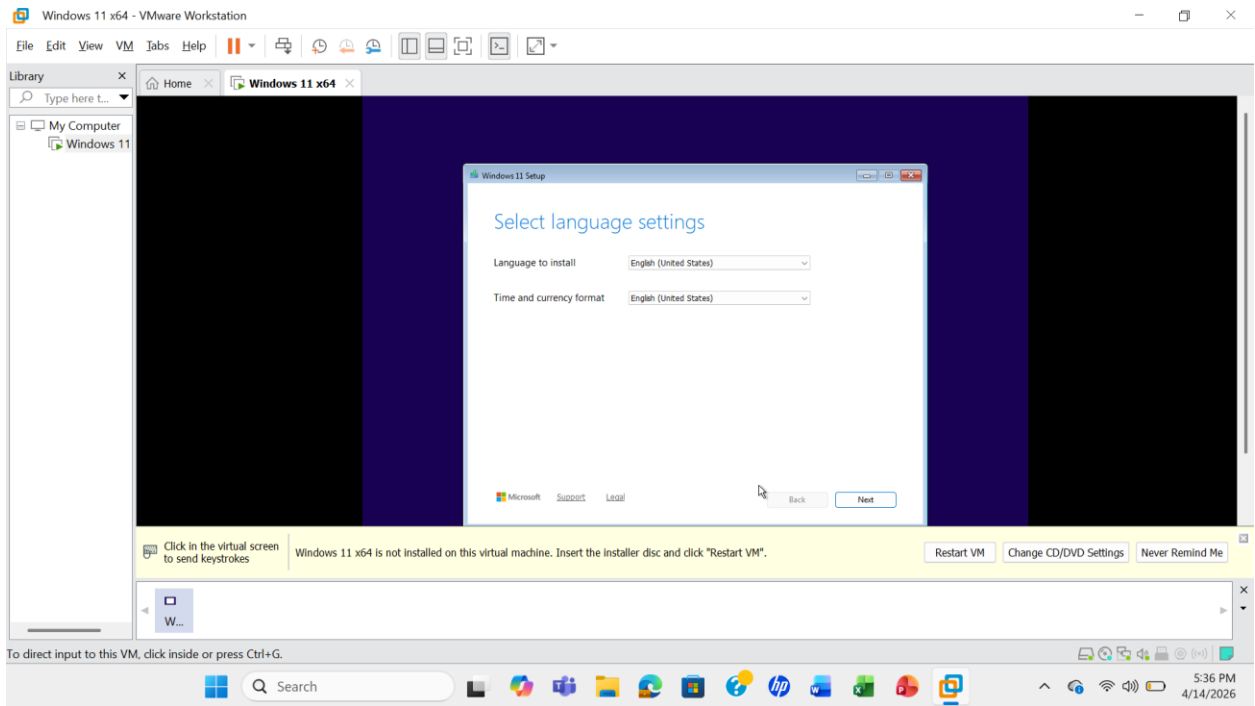




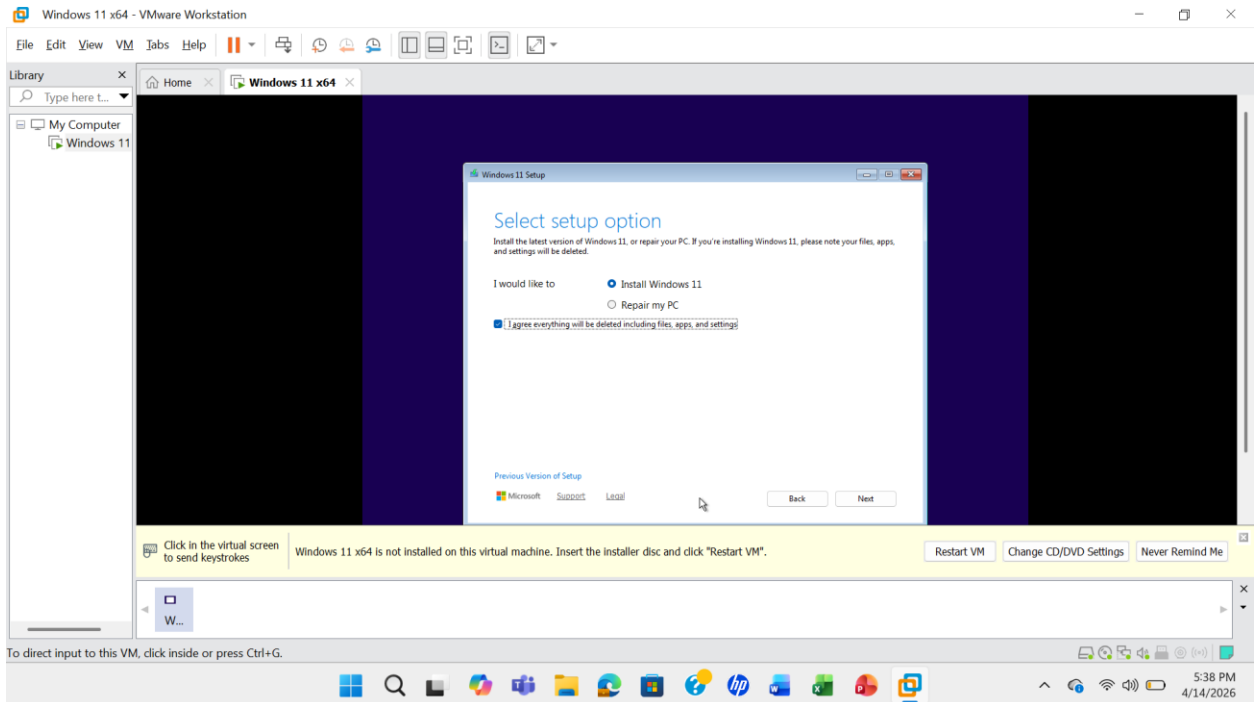
9) After these steps, the virtual machine is created and the following window will appear.



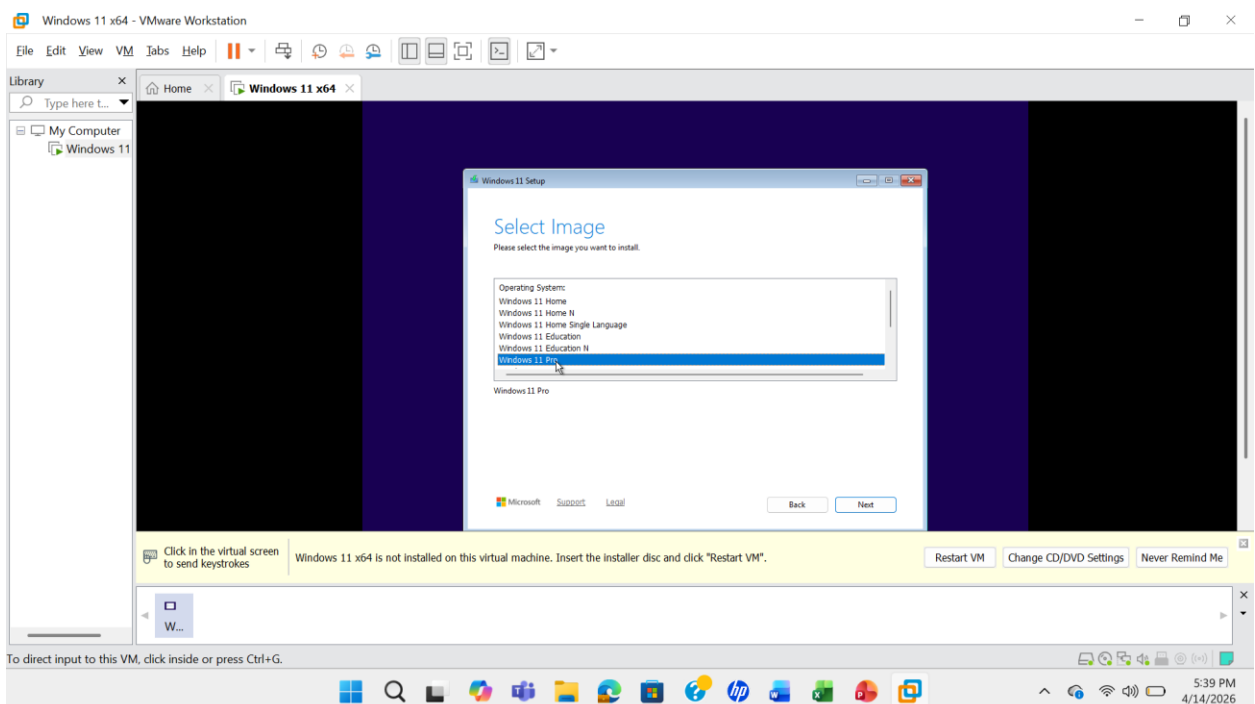
- 10) Select the power on this virtual machine and choose the language to install and time and currency format and also the keyboard/input method.

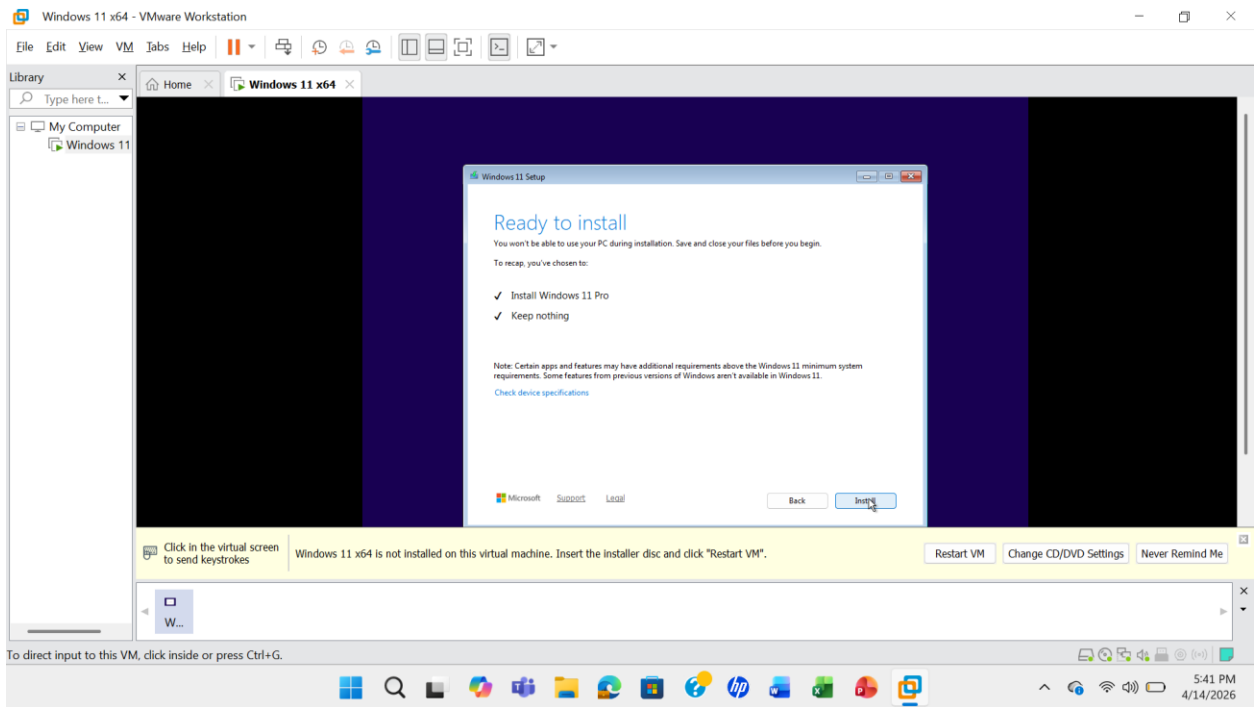
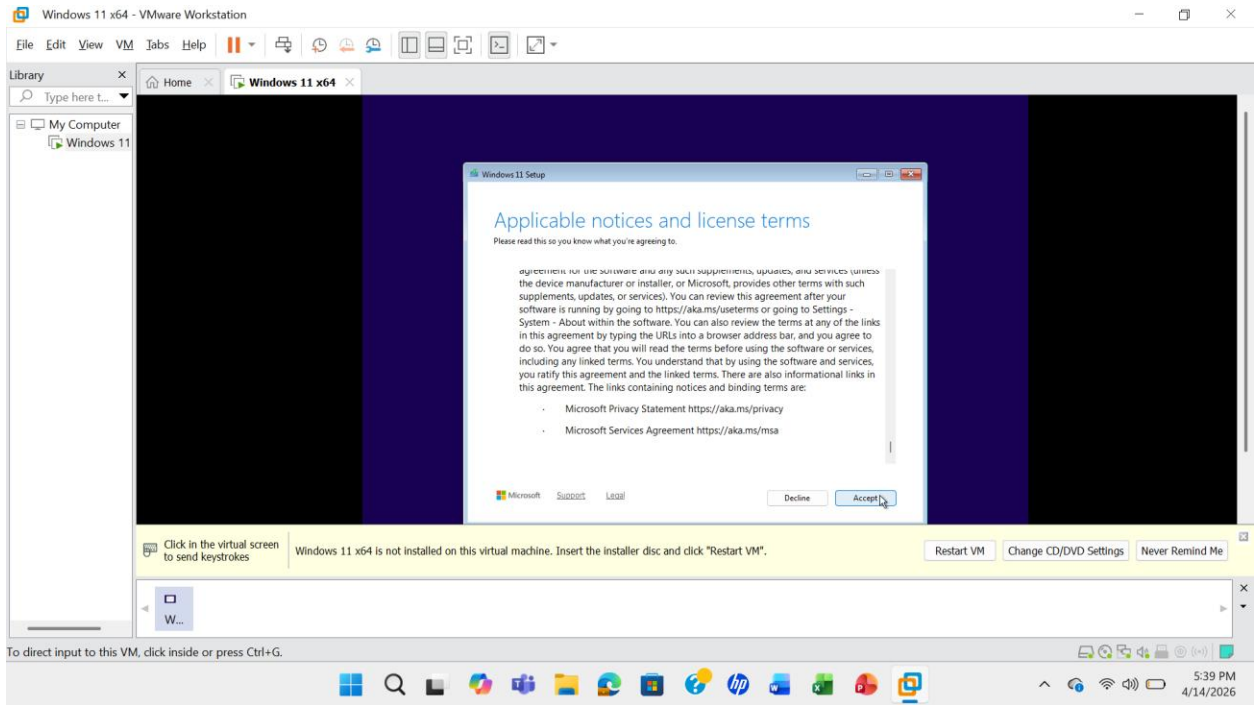


11) We selected the setup option to install the latest version of Window 11.

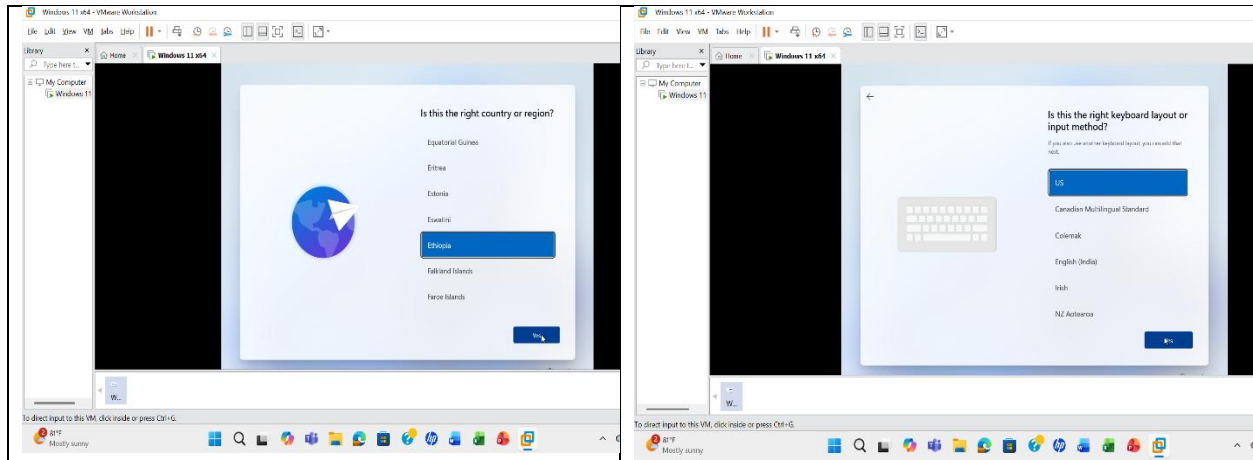


12) Select the preferred Windows 11 edition and accept the Microsoft license terms to proceed with the installation and also, we can select the location to install Window 11.

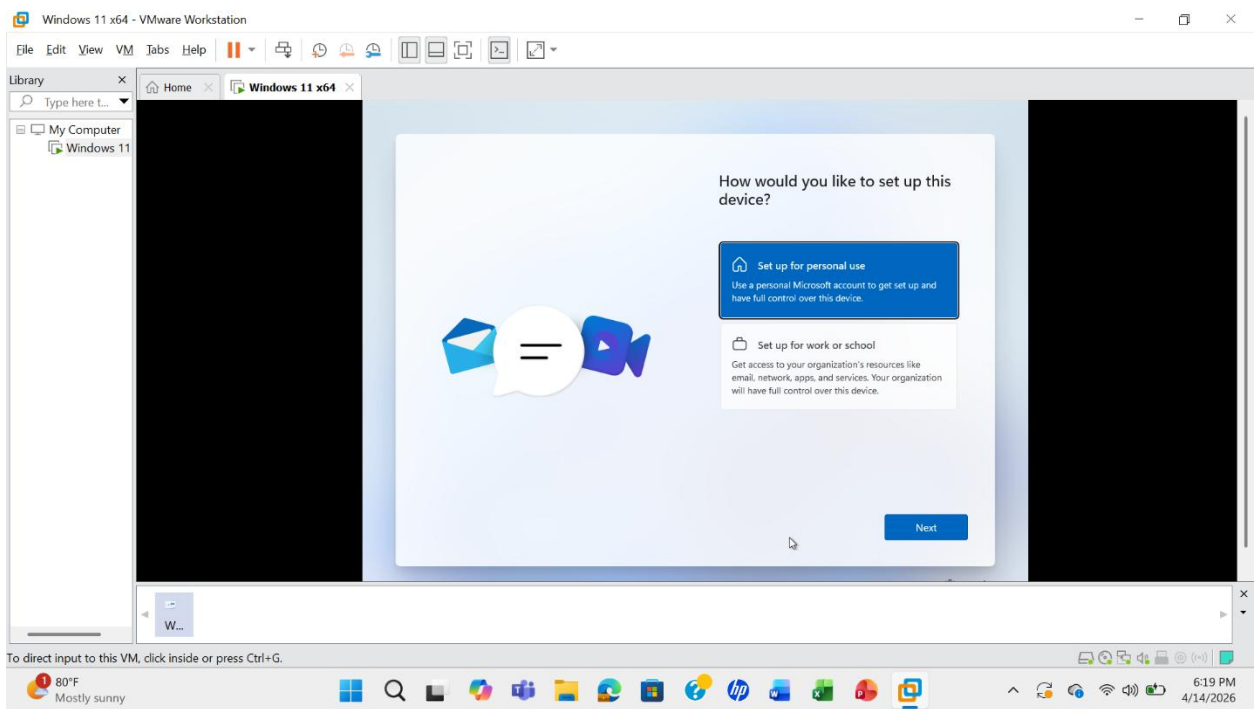


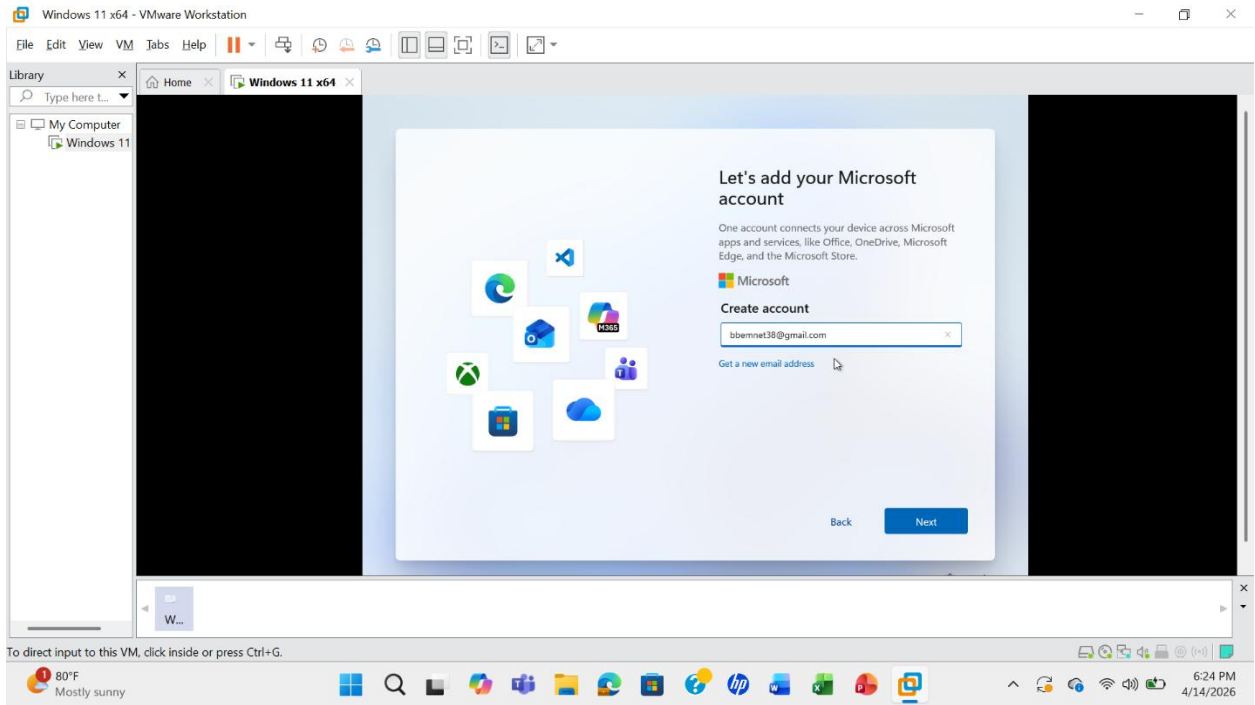


13) After the installation is done, we chose the country/region and the key board layout.

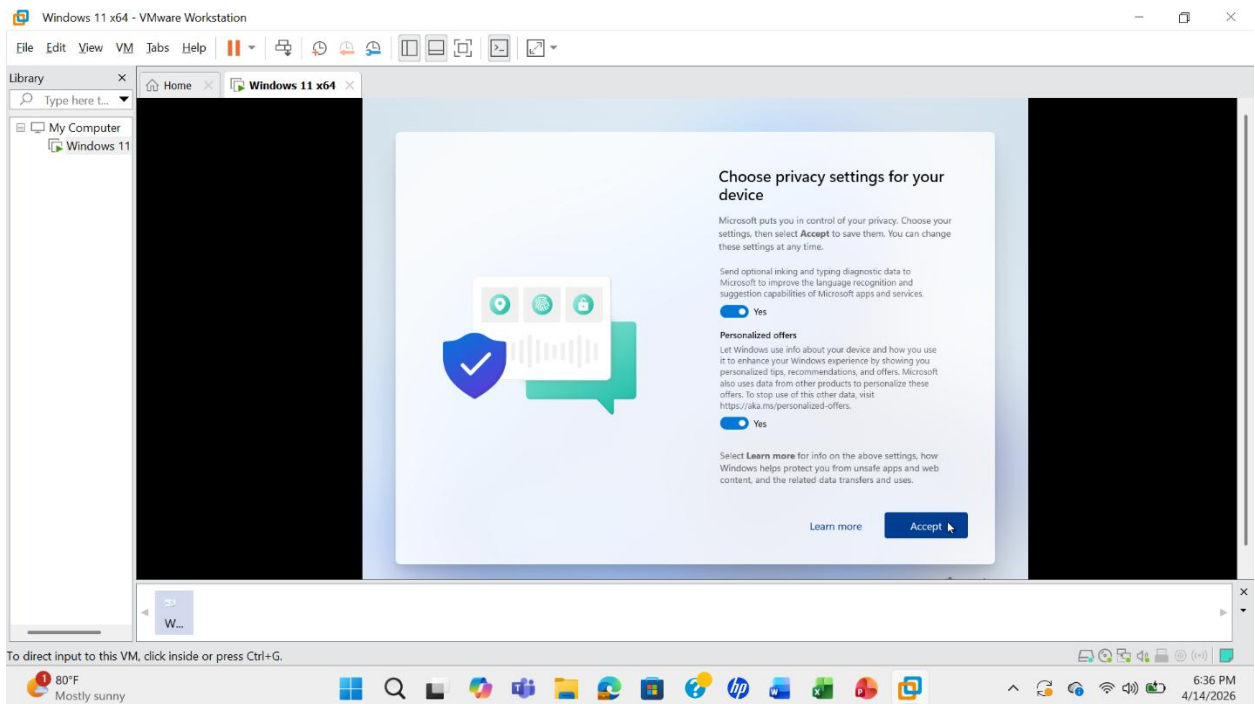


14) Once Window 11 boots into the set-up process, we set up for the personal use since we are setting up the device for ourselves and we will sign in with our Microsoft account.

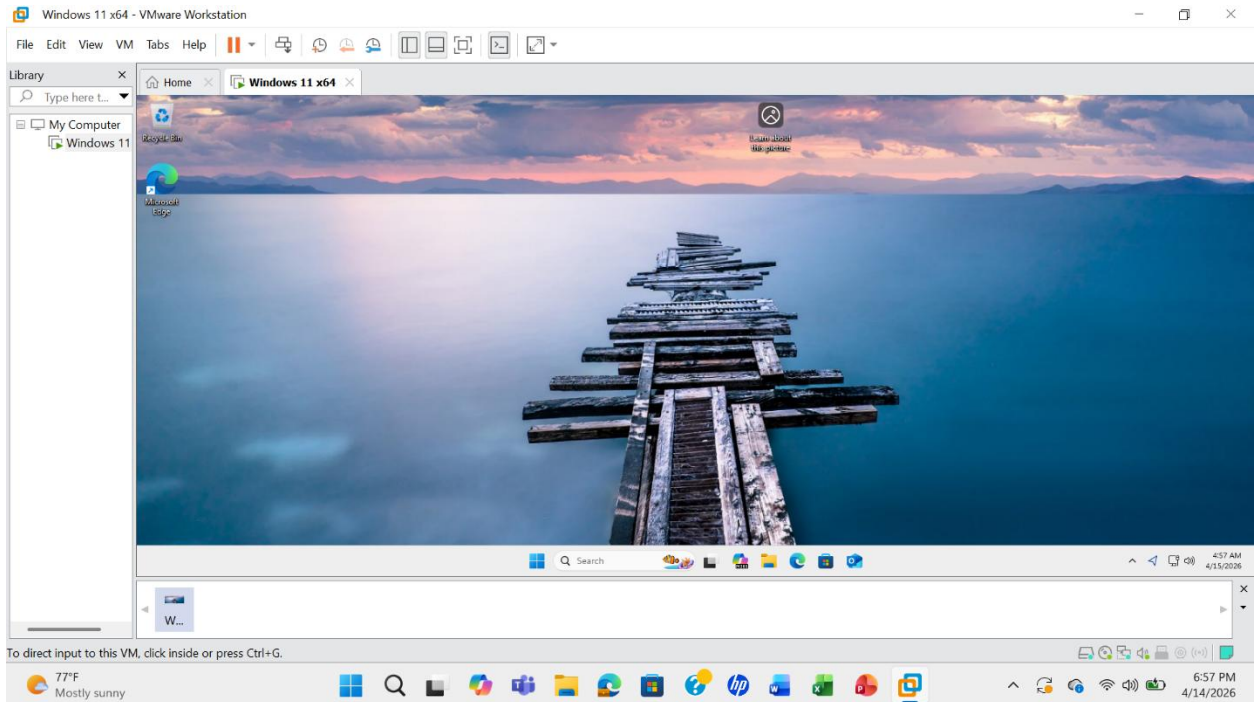




- 15) Next, the “choose privacy setting for your device” screen presented and for a standard installation, we leave both settings as they are and simply clicked the accept button to continue.



16) The installation of Windows 11 on VMware is now complete, and the system is ready to use.



5. PROBLEMS FACED AND SOLUTIONS

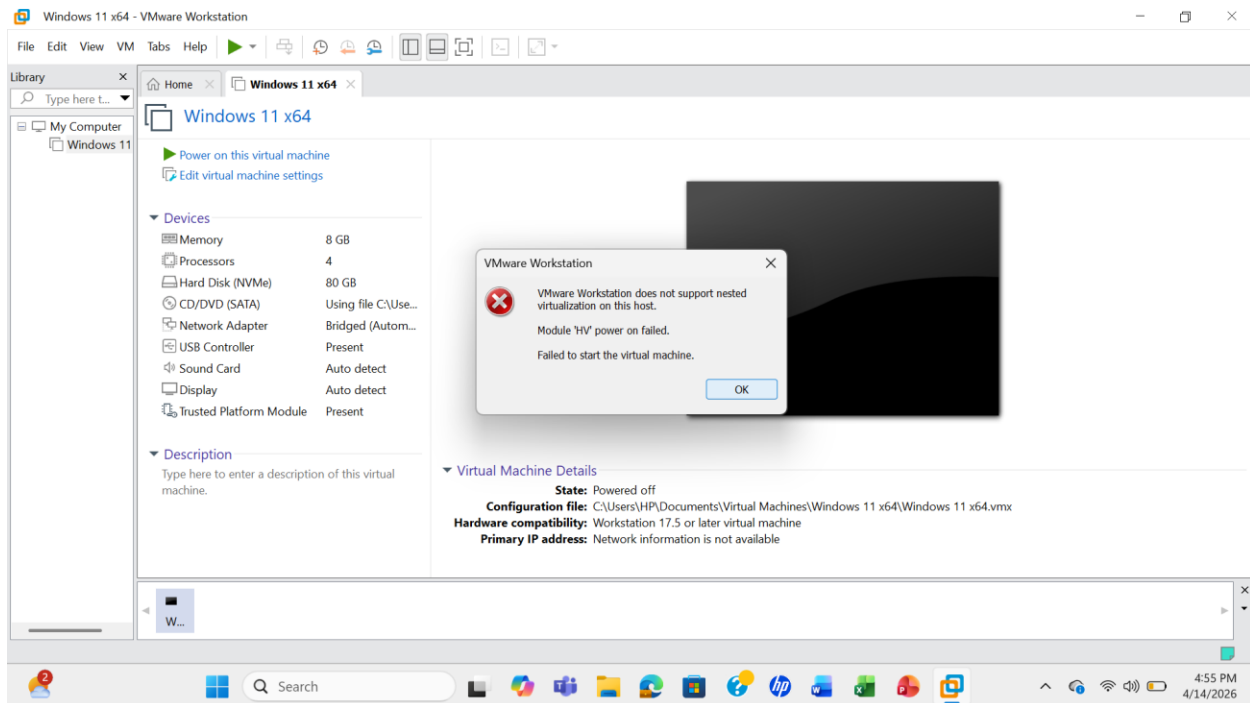
- Input device malfunctioning

When we run Windows 11 on VMware Workstation Player, the keyboard or mouse don't work properly inside the virtual machine.

Solution: The thing that we should do to fix input device problems in Windows 11 on VMware Workstation Player is to click inside the virtual machine window to make it active, and then press Ctrl + Alt to capture the keyboard and mouse correctly.

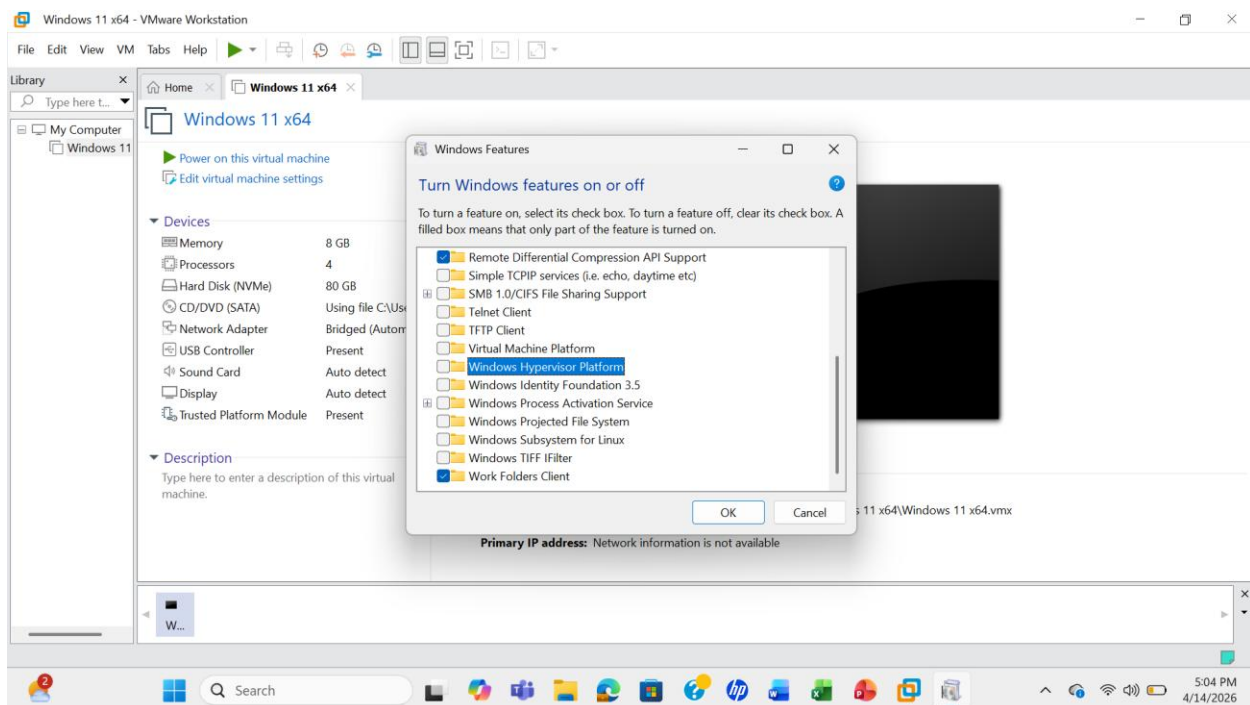
- Windows Hyper-V blocking VMware workstation

When creating the virtual machine, the Windows Hyper-V feature is activated and did not allow the VMware to function properly as shown below.

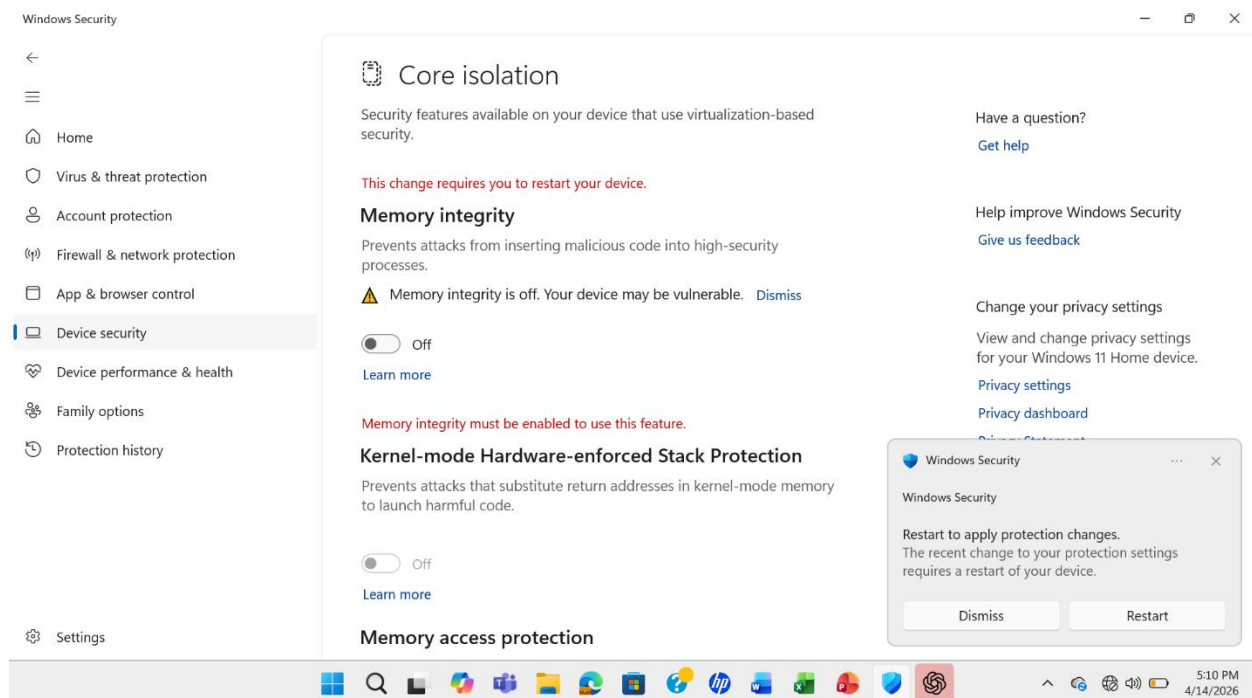
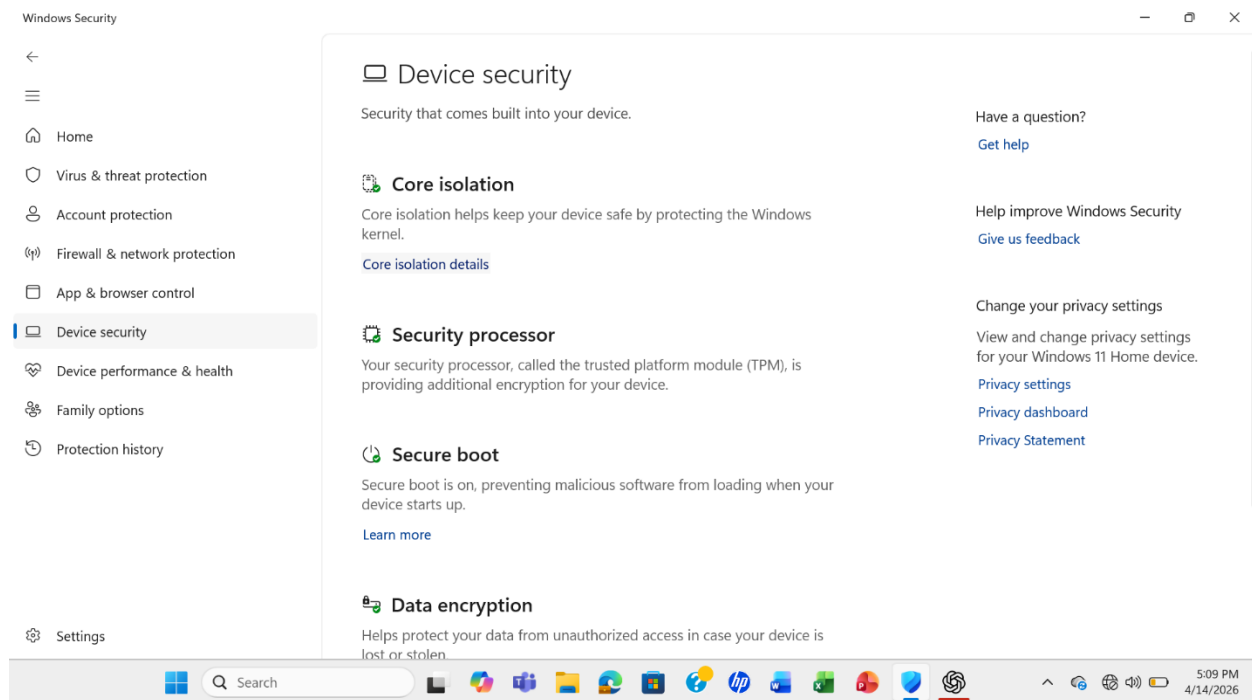


Solution: Hence changes in the system's configurations become essential to continue. In order to solve this issue, we followed the next steps.

Step 1: Disable Window features like Hyper-V, Window hypervisor platform, virtual machine platform, and Windows sandbox if they present and restart the PC.



Step 2: Disable Memory integrity from Windows security to prevent conflicts with VMware workstation and restart again.



6. FILE SUPPORT SYSTEMS

Windows 11 allows different file systems for storing data based on the need and the ability of electronic hardware tools to function together.

- NTFS (New Technology File System): Although several choices are available, NTFS remains the primary file system for this operating system because it provides high speed, strong safety characteristics for individuals who use the computer, and effective data management during every operation.
- FAT32 (File Allocation Table 32): Support for older hardware items such as USB drives is maintained by Windows 11 using FAT32. Even though FAT32 is helpful for old tools, a constraint of 4 GB for every single document remains a difficult situation within FAT32.
- exFAT (Extended File Allocation Table): exFAT is a superior choice for current external storage devices because it manages massive data sizes without the small constraints found in FAT32.

Other file systems like ext4, Btrfs, and ZFS that Linux uses are not included in Windows 11 naturally. Access to these Linux file systems is only achieved when individuals utilize extra utility programs or the subsystem known as WSL.

Apple file systems such as HFS+ and APFS are not supported by the operating system directly and require software from external vendors for connection.

7. ADVANTAGE AND DISADVANTAGE

➤ ADVANTAGES

- Modern and Attractive Interface: People get a high-quality experience when Windows 11 is installed because a tidy visual appearance is provided and the taskbar was redesigned.
- Better Multitasking: Snap Layouts and Snap Groups are used to arrange many windows at the same time and help users to stay organized and allow easier management when people have multiple tasks to complete.
- Improved Performance: The computer starts quickly because memory is handled with high efficiency by Windows 11 and power duration lasts longer for the user because energy is saved.
- Enhanced Security Safety: tools like TPM 2.0 are included in Windows 11 to protect against digital dangers and hardware-based guards are utilized to stop attacks from the network.
- Support for Modern Devices Navigation: using fingers on the screen is possible because Windows 11 was made for tablets. The software functions well on new devices which have touch abilities.

➤ **DISADVANTAGES**

- **High System Requirements:** Specific hardware like the TPM 2.0 chip is required by Windows 11 and older computing machines cannot use this version because the requirements are too high.
- **Compatibility Issues:** Previous software or drivers might not work with Windows 11 until updates are finished because compatibility problems are noticed when old programs are opened.
- **Learning Curve and Visual changes:** are introduced by Windows 11 for the benefit of users. Because the layout is new, people who use computers might feel confused and need time to understand how Windows 11 operates.
- **Limited Customization:** Limitations in changing the screen appearance are present and the taskbar cannot be moved to different sides of the screen by the user. Because of this, it offers less freedom for those who want to change settings.

8. CONCLUSION

Windows 11 stands as a contemporary computing platform created to give better speed, stronger protection, and a more pleasant experience for the person utilizing the system compared to older versions.

The setup of Windows 11 needs specific physical parts and software conditions like TPM 2.0, Secure Boot, enough memory, and matching hardware engines so that the technology remains steady and fast.

In conclusion, Windows 11 serves as a modern and safe operating system for many people. Restrictions exist which depend on the hardware power of the device and the age of the machine and due to this the usefulness of Windows 11 changes based on user needs.

9. FUTURE OUTLOOK AND RECOMMENDATION

➤ FUTURE OUTLOOK

The Microsoft business organization projects a path for Windows 11 that involves constant growth and change. The business organization continues to deeply combine computerized intelligence, online data processing, and self-acting processes within the digital control software.

Previous changes like the large update from April help Windows 11 function with high efficiency because it extends the existence period of old physical machine parts.

Personalized choices are provided by the digital control software for better use. Since many working experts and individuals in learning stay in remote locations for tasks or education, Windows 11 remains a central digital control software in this mixed environment.

➤ RECOMMENDATION

This suggestion involves upgrading to Windows 11 only if a current PC lacks the power to handle the transition smoothly in the long time ahead.

The Windows 11 digital control software specifically serves individuals in learning and working experts because it contains features that increase working output.

People with old physical machine parts should prioritize hardware improvements before the latest digital control software is installed. After the hardware is ready, Windows 11 must be kept on the newest version to ensure protection and best performance and consistent updates maintain the health of the physical machine parts.

10. VIRTUALIZATION IN MODERN OPERATING SYSTEM

➤ What is Virtualization?

The creation of a digital representation for computing assets like physical equipment platforms, data preservation units, or operating frameworks defines Virtualization. Many experts believe that Virtualization functions by allowing a single electronic processing device to mimic many computing machines simultaneously instead of employing distinct physical machines for various duties.

Every simulated space is identified as a virtual machine (VM). Such virtual machines are kept in separate areas from each other because any alterations or difficulties within one VM are prevented from impacting the others or the primary framework by the software logic.

Virtualization is extensively applied within operating frameworks, network-based processing, the examination of digital programs, and instruction in this modern era. It has been observed that individuals are able to set up and operate frameworks like Windows 11 inside a digital space. Utilities like VMware Workstation Player permit this action without changing the actual electronic processing device of the user.

➤ Why is Virtualization Important?

The importance of Virtualization is seen because various benefits exist that cause electronic processing to become more productive and adaptable.

- A primary advantage is the maximizing of operational assets. Multiple frameworks are permitted to function on one solitary machine through the use of Virtualization.
- Hardware expenses and physical room are reduced when this method is used.
- Another vital purpose involves the examination and creation of digital programs. Creators of digital programs are able to examine different operating frameworks, application tools, and revisions in a secure digital space when Virtualization is utilized, without causing danger to the primary framework.
- If a malfunction occurs, the virtual machine is easily restored or erased by the user.
- Learners in educational institutions also gain benefits from Virtualization during their instruction. These learners can gain experience in setting up different operating frameworks while they do not require costly physical equipment.
- Commercial organizations apply Virtualization within network host machines to handle many network hosts with efficiency because they seek to grow their capacity and lower the cost of upkeep.
- Because virtual machines are kept away from the primary framework, the host framework remains protected even when a virtual machine is harmed by malicious code or stops working. Such isolation is viewed as the primary defense for the main electronic processing device.

➤ **How Does Virtualization Work?**

Hypervisor, a specialized software layer, makes numerous virtual machines share number of physical computers VM enables various virtual machine shares single summit. A hypervisor takes responsibility for creating, managing and running the virtual machines. It divides the computer hardware resources such as CPU RAM storage, and network access among the virtual machines.

The hypervisor gives a slice of the physical resources for every virtual machine, when a user creates a virtual machine. It allocates XGB RAM, Y CPU cores and Z GB of storage space and it downloads a virtual machine with Windows 11, then runs it as if it were its own computer.

It is the hypervisor which makes sure that all VMs are separate and do not impact one another. It also handles communication between the VMs and the physical hardware and ensures smooth performance and stability.